

Replication Handbook

Replication package

Include immutable study protocol, contracts, hypotheses, analysis plan, data dictionary, raw and processed data, event schema, code, environment lockfile, instrument configuration, stimuli, exclusions, deviations, and results.

Replication types

Direct: same operationalization and target population. Close: controlled change in setting or implementation. Conceptual: different operationalization of the same canonical construct. Transport: new population, task, projection, or context with explicit invariance tests.

Required checks

Reproduce segmentation; verify event-to-behavior attribution; recalculate derived measures; rerun model evaluation; compare uncertainty; test robustness to alternate coding; disclose deviations.

Failure interpretation

A failed replication may challenge instrumentation, operationalization, scope, transport assumptions, or theory. Do not treat it automatically as theoretical falsification.